

you save bills too. Apart from technical reasons, price trend and servicing options are helping buyers to choose UPS and inverters easily."

Opportunities for organised and unorganised sectors

The market for inverters and UPS in India is characterised by a dominant presence of the unorganised sector, informs Mathew. However, the market for these products in India is extensive enough for both organised and unorganised sectors to co-exist, since they essentially cater to different sets of consumers.

He says, "While the unorganised sector caters to the price-sensitive consumers, organised players mostly target the brand-conscious consumers. However, the low prices offered by the unorganised players are now compelling the organised players to reduce their prices to effectively maintain their competitive edge in the market."

Although Sanghvi believes organised players are some of the largest manufacturers in the inverter industry in India and worldwide, he says, "There is a massive potential for growth for both the sectors."

Effect of grid parity on this industry

Today, the cost of power is continuously rising and, at the same time, the cost of buying a solar solution is going down. As manufacturers in this industry increase, so does the awareness of solar causing the prices to go down.

Sanghvi informs, "If we look at the last five years and consider per watt cost, it must have gone down at least by 50 per cent, if not more. And the trend is, the cost is continuously going down but our power requirement and power price is going up. So the situation is coming to something called grid parity."

Explaining grid parity, he adds, "It means the power we pay for and the power we extract, in terms of capital expenditure (CAPEX) and operational expenditure (OPEX), respectively, will be the same because of rising prices of fossil

fuels and traditional methods of supplying power, and the fact that the price of photovoltaic module, which are made from the abundantly-available, easily-extractable silicon, is going down."

This point of intersection is called grid parity and hence, in the future, the solar industry would definitely be in a very strong position to expand rapidly. Once grid parity is realised, the payback period comes down drastically.

Suppose you use a generator, have a grid tie inverter extracting sunlight and using it as power, your payback period can come down to as low as 1.6 years. This will result in making the purchase decision an easier one.

Market and business trends

Extensive research by market research agencies on power backup systems market across rural and urban sectors brings up a number of trends. Suppliers operating in the inverter industry are exploring contract manufacturing opportunities.

Mathew says, "This arrangement benefits both the service provider and the hiring firm as each can focus on his core competencies. Moreover, players operating in the UPS market are increasingly focussing on selling their products through large retail formats and on strengthening their channel sales."

He further adds, "UPS units of up to 600VA are being sold through retail channels while large configurations that require technical support are mostly being sold through channel partners."

Shifting focus to the inverter segment, previously, for hard-core backup power, people used gensets. Due to rising cost and storage challenges of diesel, people are starting to shift from gensets to power inverters. Although, Das notes, "In some of the small office/home office (SOHO) and SME segments, diesel gensets are not completely being discarded but their usage is slowly being reduced; they are complementing the power inverters when they are completely discharged."

A lot of people today want to be

independent of the grid. So in the inverter segment, people are exploring solar inverters today. "The market trend is moving towards solar. Backup solutions given through solar are the most sought after," says Sanghvi. Ideally, people would install a solar system with batteries so that it not only serves their power needs during the day but also gives them a small relief in their power needs during night.

Another important business trend is an increasing demand for SOHO/high-end inverters as a lot of users are looking for uninterrupted power backup for commercial activities which saves them huge money in comparison to generators.

Manoj Jain, vice president, Microtek International Pvt Ltd shares, "Due to errant power conditions, more and more industries and SOHO segment users are shifting from generators to inverters. This is because running costs of inverters are very low and these are more convenient."

Specifically talking about the India market, Chola says, "We see the emerging requirement for inverters with 2kV and high-frequency inverters." Furthermore, the popularity of solar technologies in both rural and urban India is driving the trend for solar hybrid inverters that combine solar panels with an inverter, to address power requirements. Finally, there is also a trend of battery management in inverter batteries.

Government policies and their effects

The Jawarhalal Nehru National Solar Mission (JNNSM) currently gives 80 per cent accelerated depreciation for companies that by a solar system. And in Maharashtra, there are no taxes for a solar power generating pack. A solar power generating pack comprises a battery and the solar panel with an inverter, which attracts zero tax.

The government is releasing schemes but, at the same time, due to lack of knowledge, the assemblers and integrators still do not know how to give you maximum benefit, informs Sanghvi. He says, "So the outside pic-